

Release Plan

Note: Release plans will not include Project or Code Management stories (i.e. UID 1-3)

Release 1

Delivery date: 5th May 2025

Total Story Points: 200

This release will focus on delivering the project's Minimum Viable Product (MVP) and, other high priority stories or stories that add quality to the foundational features of the computer vision system.

1. Model Validation

In order to implement detection for the computer vision system, inferences models need to be created and trained on datasets so that the system can learn then detected various objects within selected images. The focus of this module is to create large datasets with varying conditions to increase confidence.

User Story ID	Story Title	Story Points
7	Ruck and Lineout Detection Dataset Training	18
17	Post-Game Offside Event Analysis	20
21	Model Validation with Diverse Test Data	18
Story Point Sub-Total:		56

2. Object Inference & Tracking

The implementation of this module will allow the computer vision to complete inference and tracking of key elements such as the ball and players, as individuals, teams, and collections (i.e. ruck and lineout).

User Story ID	Story Title	Story Points
5	Ball Tracking During Play	15
6	Player Team Classification by Jersey Colour	15
22	Robust Handling of Edge Cases in Detection	10
Story Point Sub-Total:		40

3. Offside Detection & Visualisation

This feature group enables the automated detection and visualisation of offside lines in rugby game footage using computer vision. By combining field line detection, coordinate mapping, and player classification, the system determines whether players are offside during key gameplay events such as rucks and lineouts. The module aligns offside lines to the field's perspective and highlights infringing players, providing clear visual feedback for coaches and analysts. This supports both tactical review and referee performance evaluation.

User Story ID	Story Title	Story Points
4	Field Line Detection and Orientation Alignment	10
10	Offside Player Movement and Tracking	28
11	Last Feet Coordinate Detection in Ruck Formations	10
12	Offside Distance Measurement and Rule Validation	10
15	Drawing Field Markings with Accurate Perspective	10
16	Entity Classification & Differentiation	8
18	Automated Offside Detection for Coaches	28
Story Point Sub-Total:		104

Release 2

Delivery date: 19th May 2025 POSTPONED FOR FUTURE DEVELOPMENT

Total Story Points: 78

This release will focus on refactoring features in Release 1 to implement further game event detection (i.e. kickoff and scrum offside) as well as storing analysis, photos and statistics after being processed by the system and reviewing it on a designated platform.

4. Extended Event Inference & Tracking

This module will enable the computer vision to detect offside during other key events such as kickoff & scrum.

User Story ID	Story Title	Story Points
---------------	-------------	--------------

8	Kickoff Detection Dataset Training	10
9	Scrum Detection Dataset Training	10
Story Point Sub-Total:		20

5. Extended Every Inference & Tracking

This module will enable the computer vision to detect offsides during other key events such as kickoff & scrum.

User Story ID	Story Title	Story Points
13	Web Application for User Interaction	18
14	Database Design for Teams and Offside Events	24
19	Missed Offsides Summary for Referee Review	8
20	Model Validation with Diverse Test Data	8
Story Point Sub-Total:		58